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Equatorial F-Region Responses to the Main Phase of the 10 May 2024 Geomagnetic Storm Over the South American Sector: Observations and Model Simulations

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Key Points:

- Extreme and simultaneous uplift of the equatorial *F* layer over Brazil and Peru at the onset of the 10 May 2024 geomagnetic superstorm
- Combined effects of large-scale traveling ionospheric disturbances and the eastward electric field on F-layer uplift and foF2 decrease over Brazil near sunset

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Abstract This study examines the equatorial ionospheric response over the South American sector during the onset of the extreme 2024 Mother's Day geomagnetic storm, which began at ∼17 UT on 10 May and reached the G5 level. Pronounced disturbances in ionospheric electrodynamics were observed under both daytime and nighttime conditions. Variations in the F2-layer critical frequency (foF2) and peak height (hmF2) were analyzed using ionosonde observations from equatorial stations at Belém and São Luís (SL) (Brazil) and Jicamarca (Peru), together with simulations performed with the SUPIM-INPE model. At the onset of the storm's main phase, a simultaneous and abrupt uplift of the F2 layer was observed at all three stations. Model results indicate that this enhancement was primarily driven by eastward undershielding electric fields, with meridional winds contributing to hmF2 variations over the Brazilian sector. Around 19:00 UT, a complex descent of the ionosphere was detected, which is attributed to the combined effects of westward overshielding, disturbance dynamo electric fields, meridional winds, F3-layer disruption, and large-scale traveling ionospheric disturbances (LSTIDs). Although the model reproduced the formation of the F3 layer over Belém and SL, its duration was overestimated compared to observations. On the other hand, its results showed a consistent trend. Near 20:00 UT, a pronounced uplift of the F layer, followed by an anomalous decrease in foF2, was observed over Brazil. In contrast, different features were observed at Jicamarca, suggesting distinct combined roles of LSTIDs and the eastward electric field in the Brazilian sector, as qualitatively confirmed by the modeling.

Plain Language Summary On 10 May 2024, Earth experienced an exceptionally strong geomagnetic storm known as the Mother's Day Superstorm. In this study, we examined how the equatorial ionosphere responded to this storm over the Brazilian and Peruvian sectors. We used data from three ionosondes and compared these observations with predictions from the ionospheric model SUPIM-INPE. When the storm started, all three stations recorded a sudden and strong upward movement of the ionosphere's F-region. The model successfully reproduced this behavior, showing that the main cause was a strong electric field generated by the storm. In Brazil, however, changes in meridional winds contributed to shape the ionospheric response, especially over SL. Our results indicate that several processes worked together to produce the complex ionospheric behavior observed: different types of storm-time electric fields, the temporary formation of an additional layer called F3, and LSTIDS toward the equator. These combined effects can explain the significant changes in the height and density of the ionosphere during this extreme event. We also highlight the significant roles of combined LSTID events and the eastward electric field in the strong rise of the F layer and foF2 decrease over Brazil near sunset times.

1. Introduction

The magnetic superstorm of May 10–11, 2024, known as the "Mother's Day Storm," represented one of the most intense events of the modern era, with Sym-H/Dst indices reaching extremely negative values. As reported by

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Spogli et al. (2024), this event highlighted not only the potential impacts on technological systems but also the urgent need to improve monitoring, forecasting, and preparedness for extreme space weather conditions.

From initial publications and reports, it became clear that the magnetosphere–ionosphere–thermosphere coupling was highly complex during this event, with distinct signatures being registered at different latitudes and longitudes during both the local day and night conditions. Tulasi Ram et al. (2024) investigated the electrodynamic disturbances that led to a strong positive ionospheric storm on May 10. They reported more than 100% increase in dayside ionospheric total electron content (TEC) during this event, which seriously affected GPS positioning and navigation. Additionally, they mentioned an HF radio blackout in the 2–12 MHz frequency band, which indicates a significant increase in photoionization in the D- and E-regions of the ionosphere over the Indian sector due to a solar flare that occurred prior to the storm event.

The dynamic variations of equatorial ionization anomaly (EIA) morphology in the American–Atlantic longitudes sector during the super geomagnetic storm of May 2024 were investigated by Aa et al. (2024). It was observed that during the main phase of the storm, the low-and mid-latitudes ionosphere underwent a positive storm, with TEC increasing by 50%–100% within only a few hours. The EIA crests showed a marked poleward displacement, expanding up to ±35° MLAT. As noted by the authors, this behavior was primarily driven by the intensified fountain effect induced by prompt penetration electric field (PPEF) in combination with enhanced ambipolar diffusion associated with transient surges in meridional winds.

Using measurements from the Swarm mission, Nayak et al. (2025) further showed that this event produced the strongest storm-time super-fountain observed during the entire lifetime of the Swarm mission (2013–present), with nearly a 500% increase in crest density compared to quiet-time conditions. During the recovery phase, however, the EIA was strongly suppressed by the disturbance dynamo electric field (DDEF), with density crests shifting toward the magnetic equator and density troughs forming at magnetic dip latitudes of approximately 10°–15°. Rout et al. (2024) showed that initially the EIA crest locations in the Northern and Southern Hemispheres along the Jicamarca meridian presented an asymmetry that gradually increased and then decreased after 22:00 UT on 10 May. The deviation in the northern EIA crest was smaller (24°–35° MLAT) compared to the deviation in the southern crest (15°–34° MLAT). In parallel, they also noted that the wind velocity at a nearby longitude over the Brazilian sector of Cachoeira Paulista (22.7°S, 45°W) exhibited a strong poleward wind (∼−180 m/s) during the evening hours on the event day, which was absent on the previous quiet day. They further observed that the wind speed began to decrease from ∼20:00 UT on 10 May and closely matched the pattern observed on the previous quiet day after 22:00 UT, suggesting that the weakening of the meridional wind may have contributed to the reduction of the asymmetry in the EIA crest locations.

Zhang et al. (2024) examined the Asian–Australian and American sectors and reported contrasting ionospheric responses between these sectors during the main phase of the storm due to a combined action of PPEF and disturbance winds. Over the American sector, the authors showed that a strong westward DDEF contributed to the dayside-negative ionospheric storm. In contrast, over the Asian–Australian sector, they reported pronounced longitudinal gradients in TEC across mid-and-low latitudes, indicating the influence of disturbance winds and variations in thermospheric composition (O/N_2 ratio).

Astafyeva, Maletckii, Ouar, et al. (2025) showed that the effects of the negative ionospheric storm persisted until the recovery phase of this superstorm due to a unique combination of factors, including a strong dayside equatorial downward drift of ∼30–37 m/s that operated continuously over South America for 6–7 hr, suppressing the development of the EIA, together with long-lasting (∼40%) depletion in the thermospheric composition ratio $\Sigma O/N_2$, which led to decreases in the absolute vertical TEC.

Zakharenkova et al. (2025) also discussed the dramatic intensification of the EIA on 10 May 2024 over South America. Additionally, they reported a remarkably tilted pattern in the rate of TEC index (ROTI) detections at mid-latitudes over South America, a feature that is markedly different from the typical signatures of equatorial plasma bubbles (EPBs). A steep TEC gradient was seen at the equatorward and poleward walls of the modified southern EIA, contributing to high TEC values within the crest's narrow area and consequently to high ROTI values in this area.

Singh et al. (2024) and Astafyeva, Maletckii, Förster, et al. (2025) highlighted a long-lasting daytime electric field penetration that intensified the vertical plasma drift over Jicamarca, giving rise to a dusk-side super plasma fountain. The eastward equatorial electrojet (EEJ) was intensified, reaching ∼260 nT on the dayside under the

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influence of the eastward PPEF. At sunset, the strong westward electric field suppressed pre-reversal enhancement, leading to downward plasma drift (∼96 m/s) at around 24:00 UT (19:00 LT).

Wu et al. (2025) investigated the ionospheric response over the magnetic equator in the Atlantic sector and identified a large-scale electron density depletion around local midnight on 10 May. Inside the depleted region, the measurements showed complex plasma dynamics, with upward ion velocities at 720 km and downward velocities at 550 km. The density profile displayed a clear double-peak structure at 21:00 LT consisting of an upper peak near 760 km, a valley around 600 km, and a lower peak at ∼440 km. Under quiet-time conditions, however, only a single typical F2 peak at ∼540 km is generally observed.

In addition to these large-scale ionospheric disturbances, Carmo et al. (2024) used a combined data set, including GNSS receivers in the Latin American sector, ionosondes, a Fabry–Pérot interferometer, and an all-sky imager to investigate the development of a super EPB observed over the American sector on 10–11 May. The EPB exhibited an anomalous westward drift of ∼140 m/s and an unusually large latitudinal extension, reaching about 36° geomagnetic latitude, corresponding to an apex height of approximately 4,500 km. The associated plasma depletion persisted for about 12 hr from 22:30 to 10:30 UT on 11 May.

Furthermore, Vichare and Bagiya (2024) examined the effects of the east–west component of the interplanetary magnetic field (IMF-By) on equatorial ionospheric electrodynamics under disturbed conditions. They proposed that when IMF-Bz is northward and IMF-By is strongly positive, an eastward PPEF may develop near the geomagnetic equator in the terminator sectors, contrary to the overshielding electric field typically expected under such conditions. These findings highlight the complexity of the electrodynamic response of the ionosphere during this geomagnetic storm.

Han et al. (2025) investigated the characteristics and hemispheric asymmetry of LSTIDs during this superstorm using TEC observations from the Asian–Australian and American sectors. Their results showed that both the amplitudes and meridional phase velocities of LSTIDs were larger in the Southern Hemisphere than in the Northern Hemisphere in the two sectors analyzed. The energy deposition patterns inferred from polar cap indices and magnetometer data also exhibited a similar hemispheric asymmetry, suggesting that differences in energy input between the hemispheres may have contributed to the asymmetric propagation of LSTIDs. In addition, further analysis indicated that horizontal neutral winds may have played an important role in modulating LSTID propagation, with Southern Hemisphere winds favoring their development, while Northern Hemisphere winds tend to suppress their propagation.

Based on the bibliographic review presented here, the May 2024 geomagnetic storm exhibits a series of extraordinary features, including strong longitudinal variability in ionospheric responses, extreme uplift of the equatorial ionosphere, exceptionally large positive TEC enhancements, and the most intense storm-time super-fountain ever observed by the Swarm mission. The event also displayed an unusual spatial configuration of ionospheric storm effects, characterized by the simultaneous occurrence of positive and negative phases at different latitudes, an extreme EIA poleward expansion accompanied by pronounced hemispheric asymmetries, exceptionally strong thermospheric winds, as well as a long-lasting suppression of the EIA during the recovery phase. Additional remarkable features included unusual vertical electron density structures, the formation of atypical ROTI patterns at mid-latitudes, and the development of an extreme super EPB. The storm was further characterized by long-lasting penetration electric fields, a strong competition between PPEF and DDEF, and a significant influence of IMF By on equatorial electrodynamics. Furthermore, LSTIDs displayed clear hemispheric asymmetry in their propagation characteristics.

Therefore, the studies cited above underscore the exceptional nature of this event and show that the May 2024 superstorm provides a valuable opportunity to advance our understanding of magnetosphere–ionosphere–thermosphere coupling under extreme space weather conditions. In this context, the present study investigates the ionospheric response over the Brazilian and Peruvian sectors during the main phase of the May 2024 extreme geomagnetic storm from a new perspective. Unlike most previous studies, which have relied mainly on multi-instrument observations, this work combines observational data with physics-based modeling to provide a more comprehensive interpretation of the ionospheric response during this remarkable event. Specifically, we employ the Sheffield University Plasmasphere–Ionosphere Model implemented at the Instituto Nacional de Pesquisas Espaciais (SUPIM-INPE) to investigate the physical processes responsible for the observed ionospheric variability over South America. To the best of our knowledge, this is the first study to apply the SUPIM-

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INPE model to examine the ionospheric electrodynamics associated with the May 2024 superstorm in the South American sector. The use of this physics-based model is particularly valuable because it allows us to assess the relative roles of disturbed electric fields and thermospheric winds in controlling key ionospheric parameters, such as foF2 and hmF2 and F3 layer formation. Our results reveal distinct significant differences in the ionospheric responses between the two regions, which are separated by only about 2 hours in local time, highlighting the strong longitudinal variability of storm-time electrodynamics over South America. This discussion contributes to a better understanding of disturbed daytime electrodynamic conditions, since most previous analyses in this sector have focused primarily on the development of ionospheric irregularities.

2. Experimental Data and Modeling

The extreme geomagnetic storm of May 2024 occurred under the influence of seven halo-type Coronal Mass Ejections (CMEs) corresponding to a complex ejecta as defined by Burlaga et al. (2001), Burlaga et al. (2002). According to Da Silva et al. (2025), a proton density of about ∼30 cm^{-3}, propagating through the interplanetary medium with a speed of ∼600 km/s, reached Earth's bow shock around 17:00 UT on May 10. This strong and abrupt increase in both velocity and density was associated with the arrival of the second Interplanetary Coronal Mass Ejection, originating from the active region AR13664 (Helioseismic Magnetic Imager onboard the Solar Dynamics Observatory). At the same time, the magnetopause standoff distance (Rmp) significantly compressed to ∼6 Earth's radius (model of Shue et al., 1998), initiating the transfer of solar wind Poynting flux into the magnetosphere as described by the Epsilon parameter of Akasofu (Perreault & Akasofu, 1978). These interplanetary conditions, associated with a complex ejecta, triggered a complex magnetosphere-ionosphere coupling, as discussed in this work.

2.1. Ionospheric Data on 10 May 2024

The ionospheric responses to the extreme geomagnetic storm of 10 May 2024 were investigated using Digisonde data from the Brazilian equatorial stations of São Luís (SL; 2.6°S, 44.2°W) and Belém (BL; 1.4°S, 48.4°W), both with LT = UT − 3 hr, and Vertical Incidence Pulsed Ionospheric Radar (VIPIR) ionosonde data from Jicamarca (JC; 12.0°S, 76.8°W), LT = UT − 5 hr. The parameters analyzed include the true heights (hF) at successive plasma frequencies (1–12 MHz) of the F-layer, the height of the F-layer peak (hmF2), and the F2 layer critical frequency (foF2).

The Digisonde is a sweeping-frequency radar that transmits electromagnetic signals vertically to the ionosphere over frequencies from 0.5 to 30 MHz. Ionospheric returns are recorded as ionograms, showing the virtual height of echoes versus frequency, and are preprocessed with the ARTIST software (Automatic Real-Time Ionogram Scaler with True Height) before manual processing using SAO-Explorer (Standard Archive Output Format). For further details on the Digisonde operation, see Reinisch et al. (2009). All ionograms used in this study were manually scaled (using the SAO-Explorer with a 6 dB threshold level) to ensure the reliability of the derived ionospheric parameters.

In the Peruvian sector, the VIPIR ionosonde was used, which employs a wide-band log-periodic antenna transmitting vertical radio pulses with a 4 kW peak power transmitter, sweeping frequencies from 0.3 to 20 MHz (Reyes et al., 2020).

2.2. SUPIM-INPE Model

To investigate the variability of the daytime ionosphere in terms of foF2 and hmF2 during the onset and main phase of the storm, we employed the physical model SUPIM-INPE. This is the first time that this storm has been modeled using SUPIM-INPE. The model solves the coupled, time-dependent equations of continuity, momentum, and energy balance for ions (O^+, H^+, He^+, n^+, $N_2{}^+$, NO^+, and $O_2{}^+$) and electrons along closed magnetic field lines. SUPIM-INPE is a modified version of SUPIM (Bailey & Balan, 1996; Bailey et al., 1993; Bailey & Sellek, 1990) developed at the National Institute for Space Research (INPE), extending the field line calculation down to 90 km instead of the original 120 km (Souza et al., 2010, 2013; Santos et al., 2016a, 2016b).

Ionospheric production is computed using the ionizing solar flux from the model developed by Richards et al. (2010) based on measurements of the Solar EUV Experiment instrument on the TIMED satellite; and from the solar EUV flux model for Aeronomic Calculations (HEUVAC, Richards et al., 2006). The model includes

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chemical reactions responsible for ion production and loss, involving both ions and neutral constituents, with neutral densities provided by the NRLMSISE-00 model (Picone et al., 2002). Ambipolar diffusion, ion–ion and ion–neutral collisions are also incorporated. To account for coupling with neutral dynamics, SUPIM-INPE relies on the Horizontal Wind Model (HWM14) for neutral wind specification (Drob et al., 2015). Thermodynamic processes, such as frictional heating, local heating and cooling, and thermal diffusion, are included, while plasma transport, perpendicular to magnetic field lines, is addressed via the $\mathbf{E} \times \mathbf{B}$ drift.

Vertical drifts from the JULIA radar were used in the model for the Peruvian sector calculations, while for Brazil, a composed vertical drift based on magnetometer (Anderson et al., 2002), Digisonde measurements, and drifts from JC data was considered. For both regions, the empirical drift model (F08, Fejer et al., 2008) was also employed to complement missing or inaccurate vertical drift data. More detail about these drifts will be given in next section.

3. Results and Discussion

3.1. Geomagnetic Storm of 10 May 2024

Figure 1 shows the variations of selected interplanetary parameters and geomagnetic indices from 12 UT to 24 UT on May 10, 2024. From top to bottom, the panels display: (a) the Sym-H index; (b) the IMF Bx and By components; (c) the IMF Bz component; (d) the solar wind velocity (Vsw); (e) the interplanetary electric field (Ey); and (f) the auroral electrojet index (AE). The intervals marked by vertically dashed lines 1, 2, and 3 correspond to periods of great interest in this work and will be discussed along the text.

The Sym-H index shows a Storm Sudden Commencement, which peaked near +100 nT at ∼17UT (vertical dashed line 1). It reached an impressive value of–518 nT at 02:14 UT on 11 May 2024 (see e.g. Astafyeva, Maletckii, Ouar, et al., 2025; Han et al., 2025; Resende et al., 2024; among others). The IMF Bx and By components fluctuated near zero before the storm, followed by pronounced rotations after 17:30 UT.

The IMF Bz was initially near zero but turned rapidly southward after ∼17:00 UT, reaching values below −15 nT. This Bz inversion caused a large undershielding electric field that strongly affected the equatorial ionosphere, as discussed in Figure 2. During the first hours of the storm, Bz also showed brief northward excursions, for example, at 17:40 UT and 18:50 UT. Solar wind velocity increased abruptly after the interplanetary shock, fluctuating between 600 and 750 km/s during the development of this storm. The interplanetary electric field Ey was predominantly positive (except at certain times, such as ∼22:30 UT), exceeding 25 mV/m at ∼18:00 UT, a condition commonly associated with geomagnetic substorms (Gonzalez & Tsurutani, 1987).

The AE index reached high values during this storm, exceeding 2,000 nT around 17:30 UT, followed by a fluctuating recovery until 19:00 UT, after which it experienced a sharp increase. During the first 3 hours of the event, strong fluctuations in AE, Bz, and Ey were observed, offering favorable conditions for the PPEF into the ionosphere. At 22:30 UT, a strong northward reversal of Bz can be noted, which was also followed by a strong reversal of the IMF By component.

3.2. Ionospheric Variability Across Equatorial South America: Brazilian and Peruvian Sector

Figure 2 presents the variability of the hmF2 parameter (thick red curve) along with frequency isolines (1–12 MHz) over JC, BL, and SL. The day before the storm began (9 May) is used as a reference and is indicated by thick blue curves. The results indicate that the equatorial ionosphere responded promptly to the eastward penetration electric field associated with the storm onset (dashed vertical line 1), as well as to the rapid south–north and north–south variations of the IMF Bz component (dashed vertical lines 2 and 3, respectively). At the Brazilian stations, significant deviations from reference day values were observed during both daytime and sunset hours. At both BL and SL, hmF2 increased by approximately 200 km between 17:00 UT and 19:00 UT. At sunset, the F layer reached altitudes higher than 700 km over these both sectors. The gray and black horizontal bars represent periods during which the ionospheric layers were beyond the detectable range as well as intervals affected by ionosonde sounding limitations.

Between 20:50 UT and 21:40 UT (green shaded area in Figure 2), some uncertainty remains in the interpretation and scaling of the ionograms over the BL station. Therefore, following the rules and conventions of the URSI Handbook of Ionogram Interpretation and Reduction (Piggott & Rawer, 1972), the qualifying letter *D* was

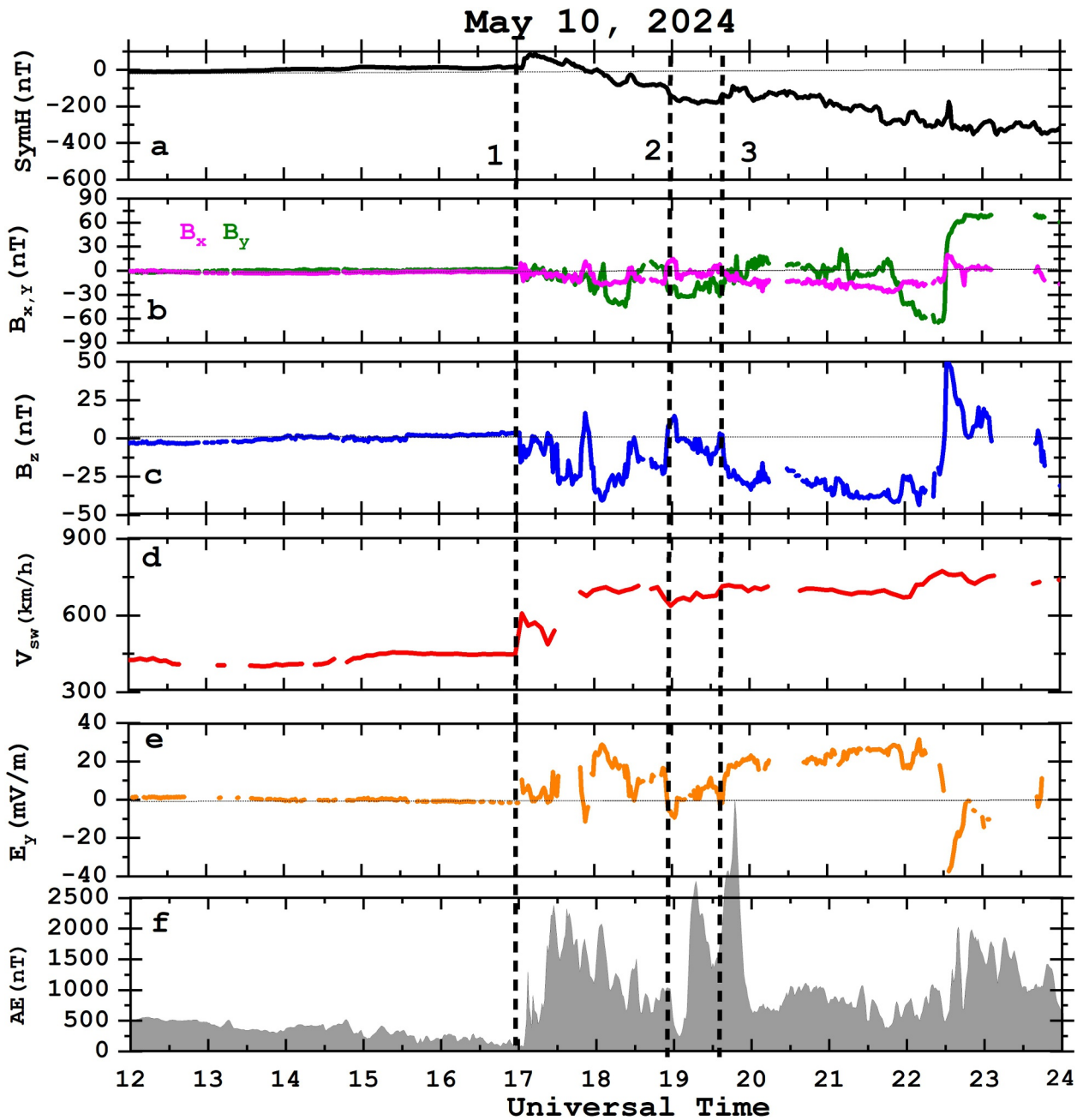


Figure 1. (a) SYM-H index, (b) interplanetary magnetic field (IMF) components Bx and By, (c) IMF Bz, (d) solar wind velocity, (e) interplanetary electric field Ey, and (f) AE index, for the period from 12:00 UT to 24:00 UT on 10 May 2024. The vertical dashed lines numbers 1 (∼17 UT), 2 (∼18:50 UT), and 3 (19:40 UT) are associated with important reversals of IMF Bz, as discussed in the text.

assigned to the corresponding hmF2 values (as well as to the foF2 values discussed in the next figures), indicating that the scaled values are affected by uncertainty and that the actual values may be higher in 5%–20% than those reported. These measurements should therefore be interpreted with appropriate caution.

Over JC, Figure 2 shows some intervals with the absence of hmF2 data after 16:30 UT (indicated by the black rectangle). Nevertheless, the isolines at higher frequencies (7–11 MHz) indicate a pronounced upward tendency of the F layer, consistent with the behavior observed in the Brazilian data. Unfortunately, during this event, the VIPIR ionosonde was configured to sound the ionosphere only up to 770 km. As a result, reliable retrievals of hmF2 and foF2 were not possible because the F-region trace was incomplete in the ionograms. Even so, the

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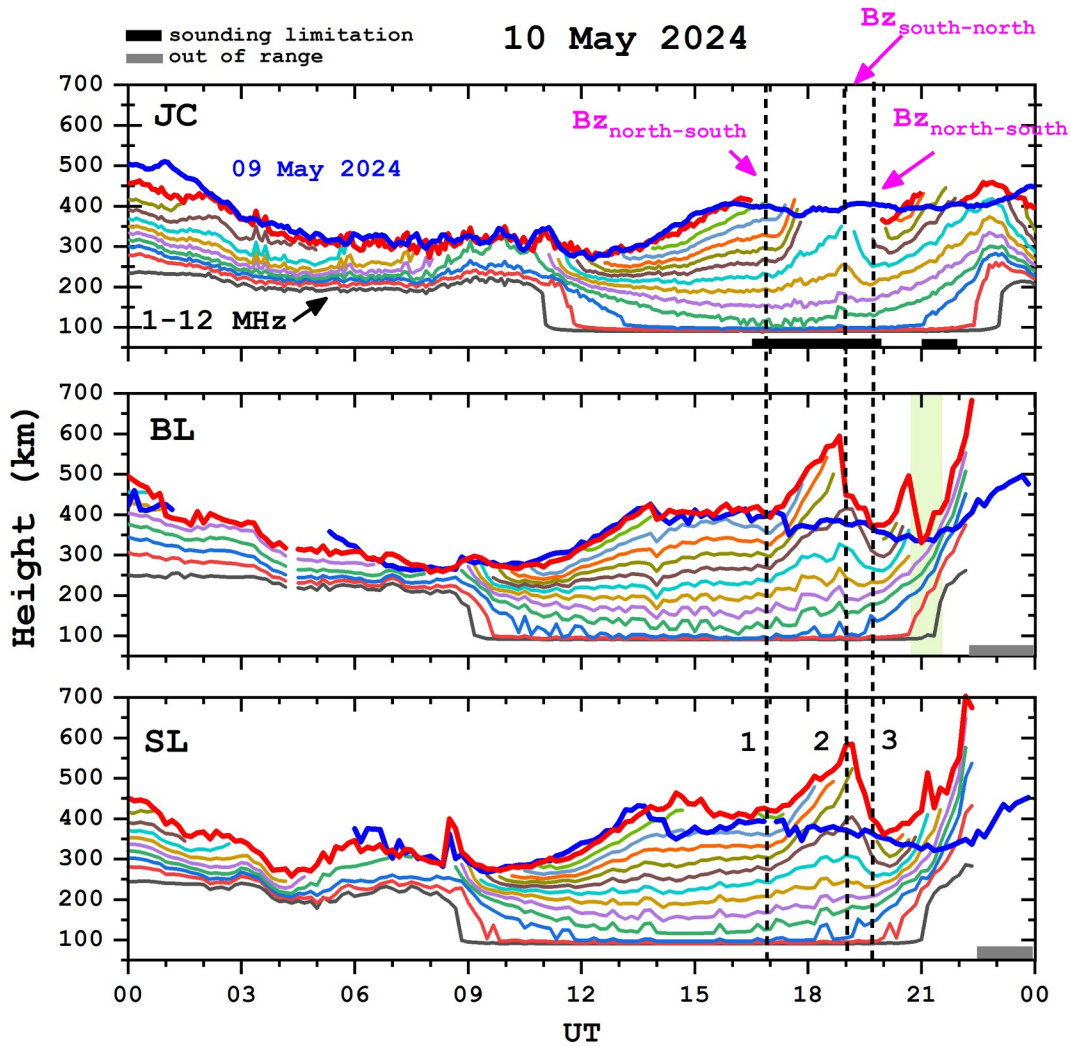


Figure 2. The ionospheric F region heights at plasma frequencies from 1 to 12 MHz, at 1 MHz intervals, and the hmF2 (thick red curve) as obtained using the SAO explorer software and the ionosonde data from Jicamarca (JC), Belém (BL), and São Luís. The hmF2 parameter for the quiet reference day (9 May) is shown as the thick blue curve. The black and gray horizontal bars indicate the time period during which the ionosphere was above the sounding limit of the equipment. The green shaded area in BL denotes intervals for which the qualifying letter *D* was assigned to the height parameters, indicating increased uncertainty in the ionogram interpretation and scaling (see text for details).

ionogram sequence in Figure 3 clearly reveals the abrupt upward motion of the F layer in this sector in response to the undershielding electric field associated with the onset of the geomagnetic storm. Furthermore, in the ionograms recorded between 17:38 UT and 17:53 UT, it is possible to observe the tendency for the F3 layer development, which in this instance likely reached altitudes exceeding the upper sounding limit (770 km). After this time, we cannot confirm whether this layer persisted or was disrupted due to the sounding limit of the equipment.

3.3. Vertical Plasma Drift Models

One of the key input parameters in SUPIM-INPE is the vertical plasma drift at equatorial stations. Therefore, it is essential to clearly describe how the drifts, equivalent to $(E \times B)/B^2$ at the magnetic equator, were derived for this event. As shown in Figure 4, the vertical drifts over JC (upper panel) were obtained from the JULIA radar measurements, while data gaps were filled using the empirical model of (Fejer et al. (2008), hereafter F08). For reference, local sunset occurs at ~22:50 UT (17:50 LT) over the Peruvian sector and ~21:10 UT (18:10 LT) over the Brazilian sector, based on standard astronomical calculations (e.g., timeanddate.com). These times are indicated in Figure 4 by the vertical cyan lines.

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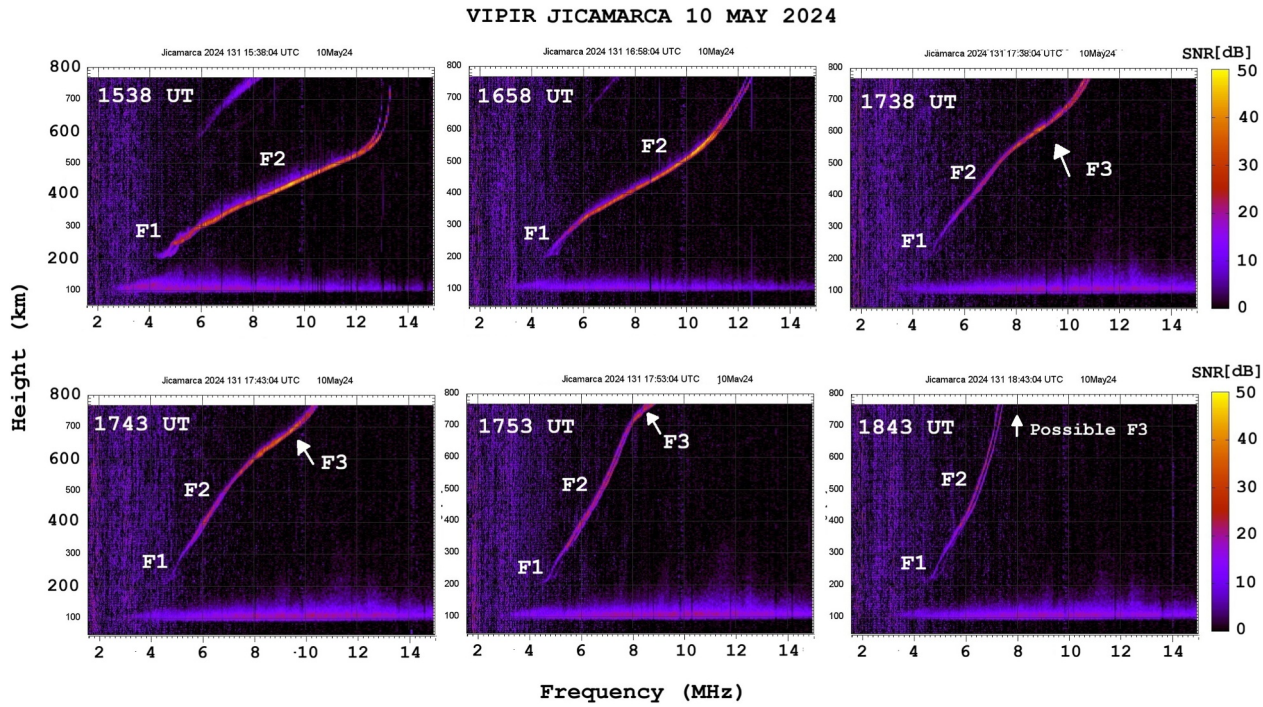


Figure 3. Vertical incidence pulsed ionospheric radar Ionograms sequence over Jicamarca on 10 May 2024 showing the presence of F1, F2 and F3 layers.

For the Brazilian sector (lower panel), two different vertical drift cases were considered. This approach was necessary because, unlike the Peruvian sector, there are no instruments across the Brazilian region that directly measure the vertical plasma drift/zonal electric field. Instead, the drift must be estimated indirectly using instruments such as ionosondes and magnetometers, both of which have important limitations. In the case of ionosondes, the drifts inferred as the time derivative of the height have limited accuracy during daytime due to the presence of the photoionization processes and strong recombination effects, which compromise the reliability of the procedure (Bittencourt & Abdu, 1981). On the other hand, drifts derived from magnetometers using the method of Anderson et al. (2002) are only available during the daytime. Moreover, as pointed out by Anderson et al. (2002), their formulation was derived from only 10 days of observations and only in the Peruvian longitude sector. Depending on the event, the plasma drifts observed in the two sectors may exhibit comparable behavior, as reported by Astafyeva et al. (2018) and in studies of the May 2024 geomagnetic storm over the American sector, or may differ substantially, as shown by Spogli et al. (2021). Therefore, the relationship between the Peruvian and Brazilian sectors is likely to depend on longitude, solar activity levels, and season conditions, although the exact degree of correspondence remains uncertain and event-dependent.

In the first case, the vertical drift was inferred using multiple independent approaches, including magnetometer-derived drifts, Digisonde observations, and the F08 model to fill gaps where magnetometer and Digisonde measurements were unavailable. Additionally, when Digisonde measurements were unavailable due to the extreme uplift of the ionospheric layer beyond the instrument detection range (22:30–24:00 UT), variations in the JC vertical drift were adopted to represent the drift behavior over Brazil. In such a condition, the temporal variation of the Jicamarca drift relative to its quiet-time pattern was used as a proxy to represent the expected variability of the equatorial electric field. This variability was then superimposed on the quiet-time drift pattern provided by the F08 model for the Brazilian sector.

In the second case, only the magnetometer drift and the JC drift as a proxy were considered. The adoption of these two cases was necessary due to the complexity of the event, which showed that the region around Brazil was influenced by disturbed electric fields, anomalous wind patterns, and the propagation of LSTIDs. Under such conditions, the estimation of the vertical plasma drift based on variations in the F-layer height may be contaminated using these processes. Therefore, the SUPIM-INPE results were generated considering these two vertical plasma drift models for this sector.

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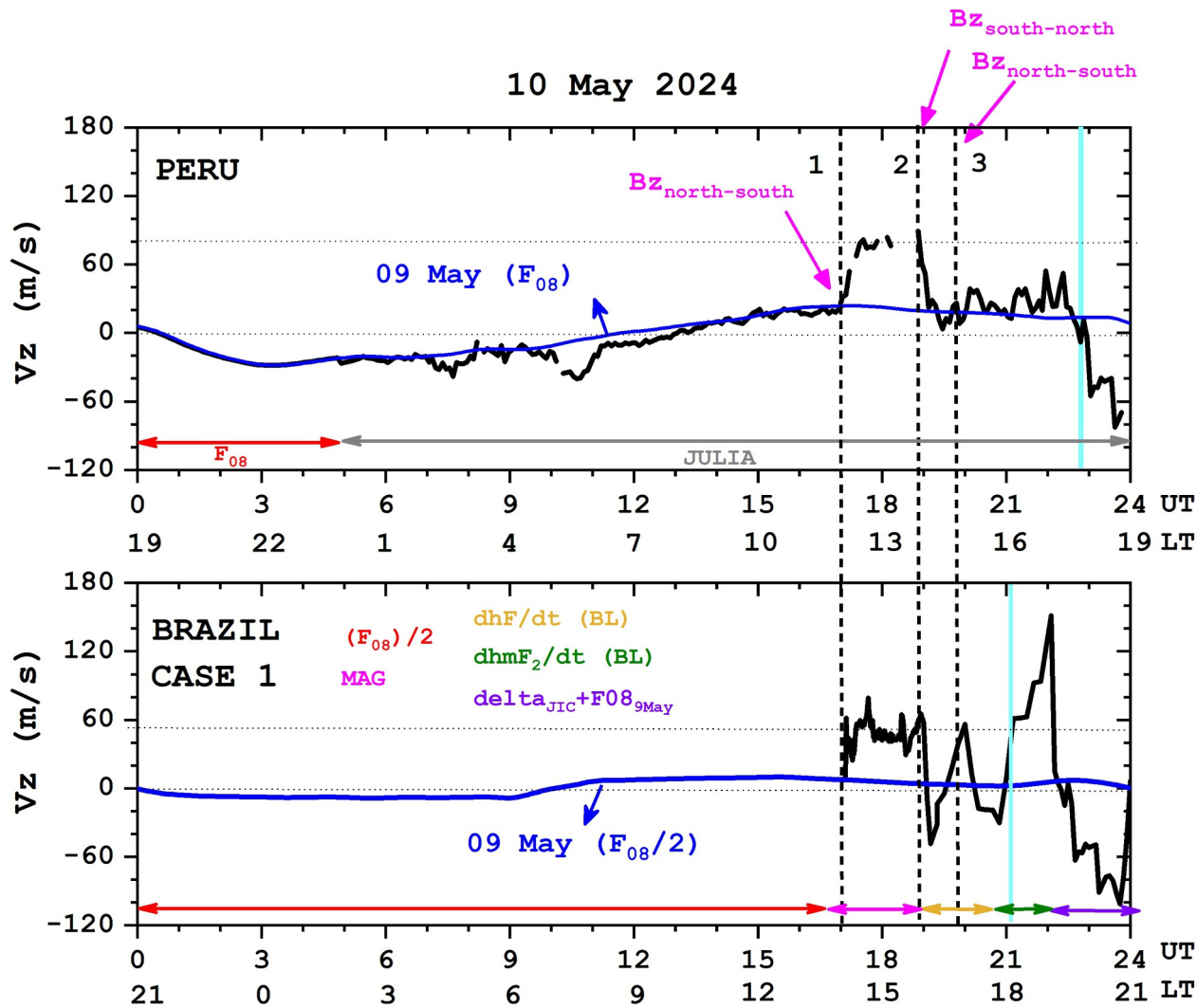


Figure 4. Vertical drifts on 10 May 2024 over Peru (upper panel) and Brazil (lower panel) used as input parameters in the SUPIM model. Blue curves indicate quiet time reference day (9 May). The colored arrows indicate the interval in which the drift values were obtained using measurements and the model. For further details, see the text. The vertical cyan lines indicate the sunset times over both longitudes.

3.3.1. Vertical Plasma Drift Model for Brazil—Case 1

The vertical drift for the first case is shown in the lower panel of Figure 4. For the interval preceding the storm onset (between 00:00 UT and 16:50 UT), the best agreement between the SUPIM-INPE results and the observed foF2 and hmF2 values was significantly improved when an adapted version of the F08 vertical drift was used as an input parameter to the model, following the approach of Nogueira et al. (2013). In the sequence, drift calculation from magnetometer data, following the methodology of Anderson et al. (2002), was performed. In this case, measurements from a station located closer to the magnetic equator, Macapá (0.04°S, 51.1°W), and another located farther away, Araguatins (5.7°S, 48.1°W) were used. This drift estimate was also employed by Astafyeva, Maletckii, Förster, et al. (2025) for the same geomagnetic storm. After several model runs with the SUPIM-INPE model, we noted that the best results for the Brazilian sector were obtained when the magnetometer-derived drift was considered only during the beginning of the storm (between 16:50 UT and 18:50 UT).

For the nighttime period, the Digisonde data were used to calculate the average vertical drift, considering the temporal variation of the real height of F-layer (hF) for 7 and 8 MHz isolines ($(dhF_{7MHz}/dt + dhF_{8MHz}/dt)/2$), and hmF2 (dhmF2/dt). The same methodology for calculating the drift using fixed plasma frequencies for the real

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height and hmF2 was also applied by Abdu et al. (2009) and Nogueira et al. (2013), respectively. However, this procedure is reliable only near sunset and during the night when the F-layer peak lies above roughly 300 km.

As shown in the isolines of Figure 2, the layer after 22:30 UT over BL (and also SL) rose to heights that exceeded the sounding limits of the instrument (~1250 km). Therefore, for this time interval, we used the JC drift variation as a reference to estimate the drift over Brazil. First, we calculated the ΔVz_{JC} parameter, defined as the disturbed drift at JC minus the quiet-time drift for 9 May obtained from the F08 model ($Vz_{10May(Julia)}$-$Vz_{9May(F08)}$). This ΔVz_{JC} component was then added to the quiet-time drift over Brazil, also derived from the F08 model ($F08_{09May} + \Delta Vz_{JC}$). A similar approach was employed by Bravo et al. (2017) to infer the SL drift from JC measurements. Therefore, we assume that the storm time electric field is similar at a wide longitude range, and the drift over the Brazilian equatorial region was estimated for the interval between 22:30 UT and 24:00 UT. Please note that, in the bottom panel of Figure 4, because F08 vertical drift is a climatological model, the curve F08/2 for 9 May overlaps that of 10 May.

As indicated by the dashed vertical lines 1 and 2 in Figure 4, the ionosphere over the equatorial Peruvian and Brazilian sectors exhibited similar responses to IMF Bz variations, producing upward and downward F-layer vertical drifts associated with eastward and westward disturbed electric fields, respectively, following the north-to-south and south-to-north IMF Bz reversals. The upward drift near 17:00 UT over Brazil increased more abruptly, indicating a faster rise in the vertical plasma motion, whereas over Peru it evolved more gradually. However, despite this sharp increase, the magnitude of the drift over the Peruvian sector was higher than that over Brazil. While in Brazil the vertical drift reached an average value of ~50 m/s, which is equivalent to an eastward electric field of ~1.25 mV/m (Fejer & Scherliess, 1995), over Peru the drift reached an impressive ~80 m/s, corresponding to ~2 mV/m. Figure 4 also shows the drift for 9 May as a representative day before the storm. Large differences are observed between quiet and disturbed conditions in both sectors.

Following the northward turning of Bz near 19:00 UT (dashed number line 2), a westward electric field developed and affected both sectors, causing an abrupt downward movement of the F layer. But, in this case, besides the effect of the overshielding electric field, it is necessary to consider the possible influence of a DDEF, since intense auroral activity had been observed in the previous hours. Additionally, as will be discussed later, the interruption of the F3 layer at this time may also have impacted the ionosphere, making the electrodynamical process at this time very complex.

After 19:00 UT, pronounced discrepancies in the vertical plasma drifts were observed between the Brazilian and Peruvian sectors. At approximately 19:40 UT (dashed line number 3), for instance, the southward turning of the IMF Bz produced only a very small perturbation in the vertical drift over JC. In sharp contrast, over the Brazilian sector (BL/SL), an initial downward motion was followed by a rapid and intense upward plasma drift, which exceeded 140 m/s by ~22:00 UT. As discussed by Carmo et al. (2024), this extreme uplift of the F layer near sunset drove a super-fountain effect, efficiently transporting plasma from the magnetic equator to latitudes of up to ~35° and resulting in pronounced plasma accumulation at the EIA crests (see also Astafyeva, Maletckii, Förster, et al., 2025; Nayak et al., 2025; Singh et al., 2024; Tulasi Ram et al., 2024).

3.3.2. SUPIM-INPE Results—Case 1

Figure 5 presents the results obtained with the SUPIM-INPE (considering the drifts shown in Figure 4), and their comparison with the observations at JC (upper), BL (middle), and SL (lower panels). The ionospheric parameters hmF2 (right panels) and foF2 (left panels) were modeled from 12:00 UT to 24:00 UT on 10 May. The VIPIR ionosonde at JC was programmed to sound the ionosphere up to 770 km; therefore, hmF2 and foF2 values at that location were used only when the sounding limit did not compromise the interpretation of these parameters. The figure also highlights the period of significant foF2 and hmF2 fluctuations (pink dashed line), which is further discussed in the keograms of Figure 7. The storm onset is marked by vertical dashed line 1.

Initial model runs were performed for the JC station (upper panels) using the original HWM14 model winds (U0). This location was selected for two main reasons: (a) the vertical plasma drift used as an input parameter is directly measured by the incoherent scatter radar at Jicamarca, providing a reliable constraint on one of the main drivers of equatorial ionospheric electrodynamics, and (b) the influence of neutral winds on the F-region dynamics is relatively small. This is due to the fact that the JC region is located near the magnetic equator and is characterized by a very small magnetic declination (close to 0°), which minimizes wind-driven effects and enhances the

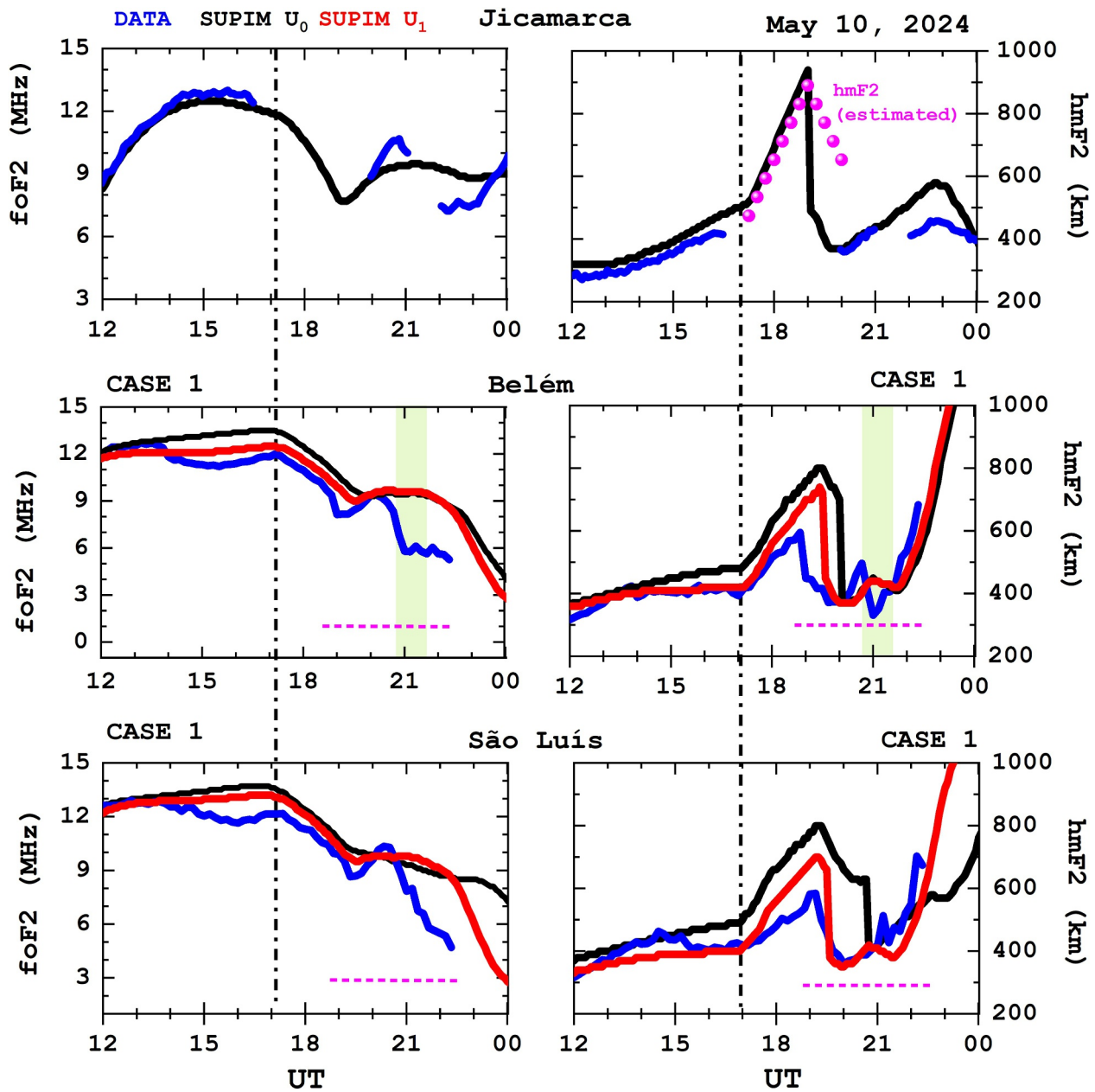


Figure 5. Observational data (blue curves) and model results (red and black curves) for May 10, considering the vertical plasma drift model of Figure 4. From top to bottom, the foF2 (left panels) and hmF2 (right panels) parameters are shown for Jicamarca, Belém, and São Luís. The black and red curves represent SUPIM-INPE results without (U_0) and with (U_1) wind correction, respectively. The subscripts 0 and 1 will refer to the unmodified and modified wind cases, respectively, as explained in the text. The pink dashed line indicates important fluctuation in foF2 and hmF2 between 18:30 UT and 22:30 UT, while the green shaded area over BL denotes intervals assigned with the URSI qualifying letter D (see text for details).

dominance of electrodynamical processes. Therefore, this configuration provides a cleaner case with fewer competing factors, allowing for a more controlled assessment of the model's performance.

In the first simulation run, the model reproduced the hmF2 parameter very well; however, it systematically overestimated foF2 values starting around 12 UT on 10 May (not shown here), that is, several hours before the onset of the geomagnetic storm. Since the vertical drift is well constrained by observations and there is no indication of significant disturbances in the neutral wind system prior to the storm, this discrepancy suggested that the solar flux used in the model was not adequately representing the effective solar ionizing input. Therefore, the solar flux was scaled by a factor of 1.8 to improve the agreement between the modeled and observed foF2 values.

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This adjustment primarily affects foF2 because the solar flux directly controls the ionization production rate and consequently the electron density. In contrast, variations in hmF2 are mainly controlled using plasma transport processes, particularly ExB drifts in the equatorial region and thermospheric winds at low and middle latitudes. Similar adjustments in the solar flux model were made by Santos et al. (2017). To maintain consistency in the simulations, the same correction was applied to the stations in the Brazilian sector. To avoid overcrowding of the figure, only the results obtained using the solar flux correction are shown for all three stations.

At about 17:00 UT, the simulation shows an abrupt rise in hmF2 over JC, reaching approximately 950 km around 19:00 UT. Although observational data (blue curves) are missing between 16:30 UT and 20:00 UT over this station, hmF2 values were estimated using the observed enhancement in the vertical drift shown in Figure 4 (upper panel). The estimate was based on an average upward drift of ∼66 m/s, derived from the arithmetic mean of the observed vertical drift (black curve in upper panel of Figure 4) between 17:00 UT and 19:00 UT. Starting from the last reliable hmF2 value of approximately 415 km (Figure 5), the resulting estimate yields a maximum hmF2 of ∼890 km at ∼19:00 UT (pink dots). Starting from the last reliable hmF2 value of approximately 415 km (Figure 5), the resulting estimate yields a maximum hmF2 of ∼890 km at ∼19:00 UT (pink dots). For the subsequent descending phase, the hmF2 values represented by the pink dots were estimated using an effective downward velocity of −66 m/s after the maximum height was reached. We progressively subtracted the corresponding height variations from the maximum estimated hmF2. This effective velocity was used only to approximate the downward evolution of the layer and should not be interpreted as a directly observed downward vertical drift. These values should therefore be regarded as approximate estimates of the F-layer evolution during the interval when direct observations were unavailable. The comparison indicates that the SUPIM-INPE model successfully reproduces the initial response of the JC ionosphere to the storm. However, some discrepancies are noted during the descending phase of hmF2, which the model represents rather abruptly.

Over the Brazilian region, the simulation was initially performed using the same solar flux and wind model used to simulate the ionosphere over JC (black curves, U_0 in Figure 5) and the vertical drift named as Case 1. It can be noted that good agreement for hmF2 and foF2 before 14:00 UT (for both BL and SL). After this time, and especially after the storm onset, discrepancies became apparent, with the model overestimating the observed frequencies. Regarding the height, the model overestimation is evident between 14:00 UT and 20:00 UT over BL and between 15:00 UT and 21:00 UT over SL. After these intervals, the model represents the hmF2 parameter over BL very well, but inconsistencies are seen over SL.

The inconsistency between the observed and modeled data for SL and BL may highlight that the wind model used for the Brazilian sector is not correct and needs some adjustments, given that the vertical drift used as input parameter in the model (deduced from magnetometer and ionosonde data) is assumed to be correct, and that the solar flux was validated for the JC region. Over JC, however, no wind adjustments were necessary, as a good agreement between the model and observations was already achieved; therefore, only U_0 curves are shown for that location.

Before discussing the results that include a correction to the wind, we first need to understand the time variation of the meridional effective wind used in the simulations for the Peruvian and Brazilian stations. Figure 6 shows that over JC, the wind was directed southward, with a smooth tendency to reverse to the north between 15:00 and 20:00 UT. After this period, a renewed shift toward southward winds is observed. Over Brazil, the initial wind (solid red and black curves, U_0) was weakly directed northward until around 16:30 UT, when a slight intensification in that direction occurred. Near 19:30 UT, the wind weakened and then reversed to southward. Simulations using these wind profiles for BL and SL were not fully satisfactory (black curves in Figure 5), requiring additional model runs to achieve results more consistent with the observational data. However, the responses were complex, sometimes improving the hmF2 predictions while degrading foF2, or vice versa, reflecting the complex interactions between winds and the ionosphere over the South American sector during this extreme geomagnetic event. A detailed discussion of this nonlinearity is beyond the scope of the present work.

After careful analysis, the wind adjustment that best reproduced the ionospheric conditions over Brazil included a parcel of 100 m/s (southward) added to the original effective winds, as indicated by the dashed red (SL) and black (BL) curves in Figure 6. In this case, the final winds (U_1) over both stations were strongly southward, with higher intensities after 19:00 UT.

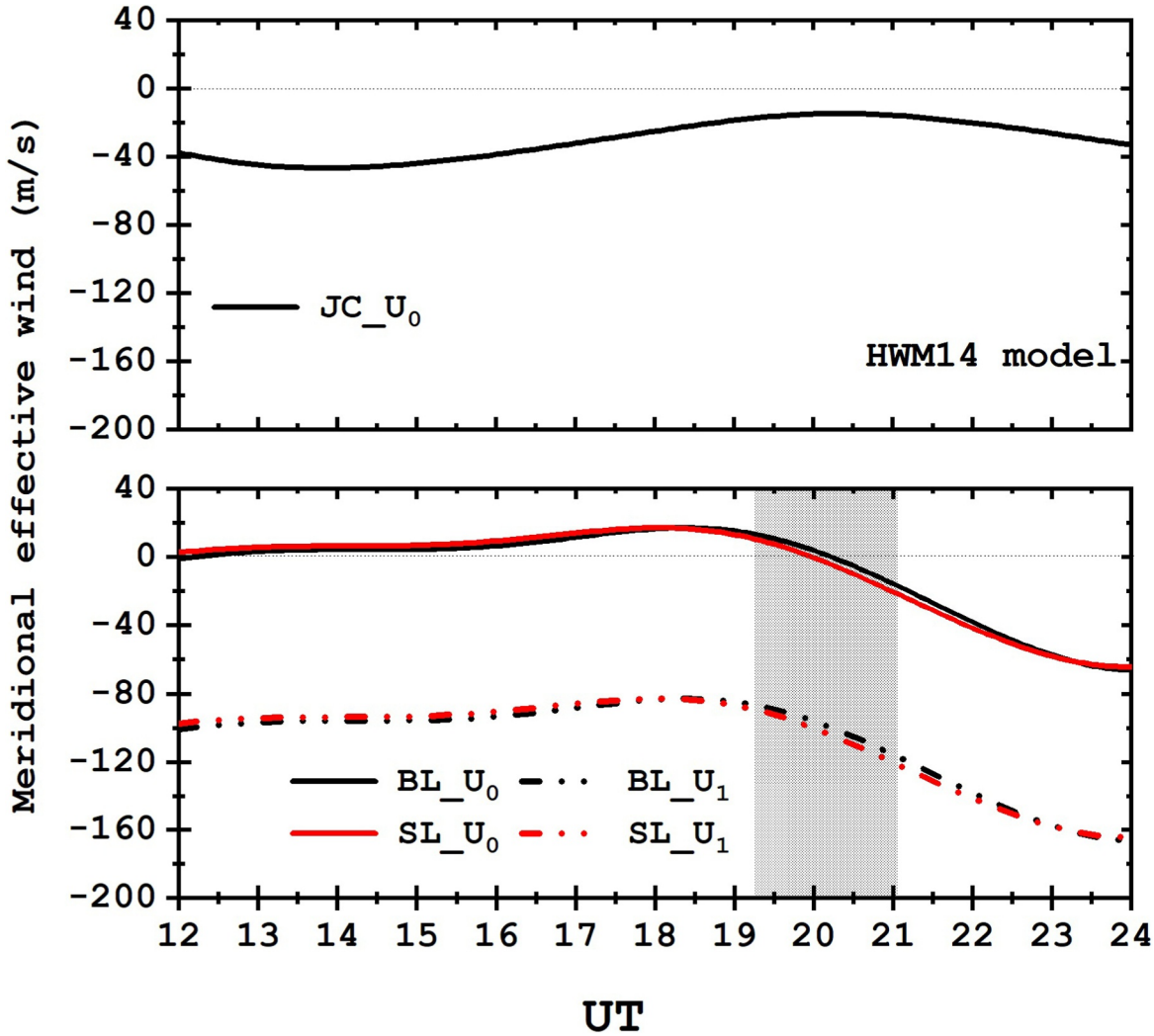


Figure 6. Initial (U_0, solid line) and modified (U_1, dashed line) meridional effective winds from the horizontal wind model model used in the simulations for the Peruvian (upper panel) and Brazilian (lower panel) regions on 10 May 2024.

According to the results including the modified wind (U_1), shown as the red curves in Figure 5, it was possible to notice that over BL, the inclusion of U_1 affected hmF2 and foF2 between 14:00 UT and 20:00 UT, contributing to bring the simulation results closer to the observations over this sector. Similar behavior is observed for SL, where the modified wind was essential to improve the descent of hmF2 after 19:00 UT and to the decrease of foF2 after 20:30 UT.

The negative meridional wind values used as input (U_1) in the present model are consistent with those reported by Rout et al. (2024) and Carmo et al. (2024) for the Brazilian sector during this geomagnetic storm, specifically at Cachoeira Paulista (22.7°S, 45°W; LT = UT–3 h), where a strong poleward wind (~–180 m/s) was observed during the afternoon of 10 May. As detailed by Aa et al. (2024), this robust poleward wind surge would greatly strengthen the ambipolar diffusion in the Southern Hemisphere via neutral drag, thereby contributing to some extent to the poleward expansion of EIA crests during this period. Thus, a combined action of thermospheric winds and electric fields likely modulated the F2-layer peak height during the initial phase of the storm, particularly over SL, which is located farther from the magnetic equator. Regarding foF2, the wind adjustments clearly improved the model performance for BL between 14:00 UT and 20:30 UT, whereas over SL, only minor changes were produced during this interval after the wind correction was applied. However, after 20:30 UT, both Brazilian stations exhibit a pronounced discrepancy between the observed and modeled foF2 values, with the model systematically overestimating the observations in both the original (U_0) and adjusted (U_1) wind cases.

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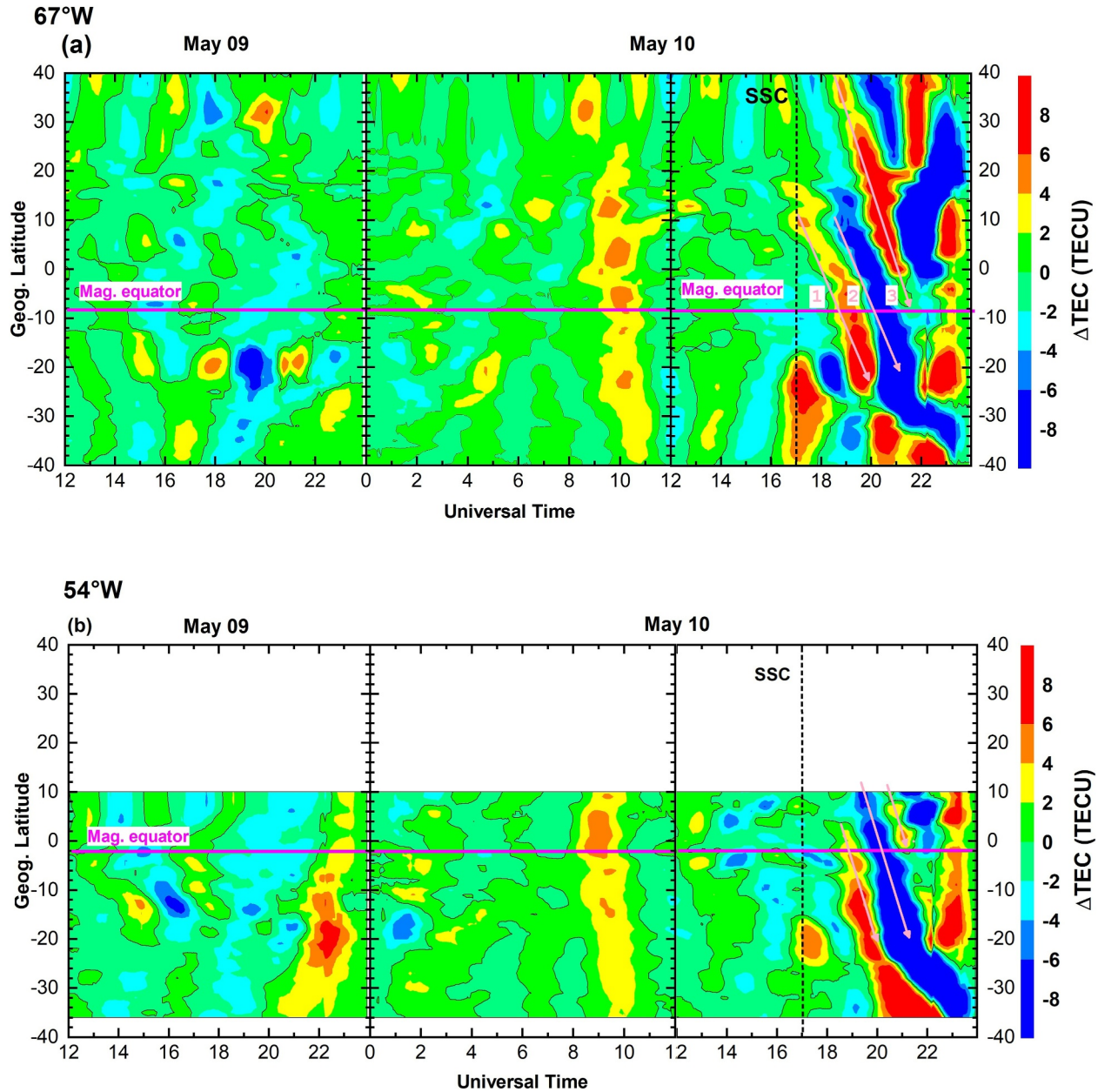


Figure 7. (a) Meridional keogram of ΔTEC along the 67°W meridian, covering the period from 12:00 UT on 9 May 2024 to the end of 10 May. (b) Same as (a) but for the 54°W meridian.

Notably, this mismatch coincides with the interval during which the vertical plasma drift over the Brazilian sector exhibited strong fluctuations, in sharp contrast to the JC station, where no comparable drift variability was observed. This contrast further underscores the distinct and highly dynamic storm-time response of the Brazilian sector and suggests that mechanisms beyond disturbed electric fields alone may have modulated the Brazilian ionosphere. Finally, these inconsistencies could be mitigated by incorporating observational wind data into the model. Although this approach would not fully resolve the discrepancies, since the model calculations are performed along magnetic field lines rather than at a single specific location, the inclusion of observational data would help to better constrain the parameters responsible for the ionospheric variability over the Brazilian sector. This improvement would provide a more realistic representation of storm-time dynamics in the region. Therefore, the incorporation of such observational wind data represents the next step of this study.

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To understand the decrease in foF2 after 20:30 UT and why the SUPIM-INPE model did not reproduce it, we examined the ionospheric behavior prior to the decrease at 20:30 UT. As shown in Figure 5, a wave-like pattern is observed shortly before the sharp decrease in foF2, between 19:00 UT and 20:30 UT, at the three locations in the analysis (SL, BL and JC). A similar oscillation in foF2 over the near-equatorial station of Araguatins, followed by a pronounced decrease in this parameter, was also reported by Fagundes et al. (2025). They noted that the O/N_2 ratio, which directly modulates ionospheric density, exhibited significant changes over the American sector, but only during the recovery phase of the storm (11 May) (see also Astafyeva, Maletckii, Ouar, et al., 2025; Evans et al., 2024). Therefore, the authors attributed that the foF2 depletion at the beginning of the event could be associated with the intense uplift of the F layer. Wu et al. (2025) also found that large density depletion developed over the magnetic equator in the Atlantic region on 10 May, along with complex plasma motions. The electron density profile showed two distinct peaks unlike the quiet day when only a single typical F2 peak is present.

It is well known that the post-sunset decrease in foF2 is associated with the reduction of ionization production (Kelley, 2009; Rishbeth & Garriott, 1969). In addition, the F-layer uplift process, driven by the prereversal enhancement (PRE) of the zonal electric field at sunset, transports plasma away from the magnetic equator toward low-latitude regions along magnetic field lines (Kelley, 2009; Scherliess & Fejer, 1999). This behavior was clearly observed over the Brazilian sector, where a pronounced uplift of the F layer at the equator occurred (~21:00 UT), followed by an increase in foF2 at the low-latitude station of São José dos Campos about 2 hours later, due to the fountain effect, as reported by Fagundes et al. (2025).

However, an important question arises: if the zonal electric field (vertical drift) used as input in the model is correct, and considering the little changes caused by the wind since hmF2 is well reproduced by SUPIM-INPE after 20:00 UT over BL, why does the model fail to capture the abrupt decrease of foF2 over the Brazilian equatorial sector?

To address this issue, we adopted the same methodology of Souza et al. (2024) to investigate the possible causes of the foF2 variability observed after 20:30 UT. For this purpose, two meridional sectors were selected for the analysis of TEC variability, as shown in Figure 7: one near the Peruvian longitude (67°W; panel a) and another near the Brazilian longitude (54°W; panel b). Figure 7 presents ΔTEC during the transition periods of 9–10 May 2024 (reference period before the onset of the storm) and 10 May 2024 (storm period). The keogram technique was applied to identify the presence of propagating structures that might account for the observed foF2 decrease. The analyses were performed along the 67°W and 54°W meridians, selected as representative transects of the western South American and Brazilian sectors, respectively. The TEC values were extracted from gridded TEC maps with a spatial resolution of $1° \times 1°$, and no additional longitudinal averaging was applied. The meridional ΔTEC keograms were obtained by subtracting the 1 hr–smoothed TEC from the 3 hr 20 min–smoothed TEC ($\Delta TEC = TEC_3h_{20} - TEC_1h_{00}$). Accordingly, positive ΔTEC values occur when the short-timescale TEC variations are weaker than the background variability represented by TEC_3h_{20}, whereas negative ΔTEC values indicate that the short-timescale fluctuations are more pronounced than the background component. This approach highlights the relative contribution of TEC variability at different temporal scales. Since traveling ionospheric disturbances can occur over a broad range of temporal scales, from medium-scale disturbances with periods of tens of minutes to large-scale disturbances with periods of several hours, the ΔTEC parameter provides a useful means of identifying wave-like structures embedded within the storm-time TEC response. For more details, see Souza et al. (2024).

The keograms in Figure 7 reveal strong wave-like structures in ΔTEC near storm onset compared to the reference day (9 May). Although there is a data gap for latitudes beyond 10°N and 38°S at 54°W, observations at 67°W suggest that similar features may also be present at 54°W. A closer inspection indicates that structure 1 is more likely an artifact generated using the filtering procedure associated with structure 2, which represents the main propagating disturbance. Because ΔTEC is defined as the difference between TEC smoothed with 3 hr 20 min and 1 hr windows, its positive and negative values reflect the relative contribution of TEC variability at different temporal scales. In the present case, the initial arrival of the propagating disturbance at low latitudes is characterized by the negative ΔTEC signature of structure 2, which is subsequently followed by alternating positive and negative variations, forming the wave-like pattern of the disturbance. Thus, the negative ΔTEC signature of structure 2 is not interpreted as the LSTID itself, but as the initial signature of the propagating disturbance. When this negative ΔTEC signature is interpreted as an enhancement of the TEC perturbation itself, its temporal evolution is consistent with the foF2 perturbations observed in Figure 5. In particular, the negative foF2 peak

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observed at BL and SL at approximately 19:00 UT coincides with the red structure in Figure 7b at the magnetic equator, while the subsequent positive foF2 peak at approximately 20:00 UT coincides with the blue structure in Figure 7b. A similar correspondence is observed for JC when comparing Figure 5 with Figure 7a, although both features occur approximately 1 hour later. Indeed, the oscillations in TEC are also observed in foF2, and when applying the keogram to foF2 ($foF2_{_3h20}$ - $foF2_{_1h00}$), the effect is the same. Based on structure 2 in Figure 7a, the estimated period (~2 hr), horizontal wavelength (~3,600 km), and phase velocity (~500 m s^{-1}) are consistent with typical characteristics of LSTIDs reported in the literature (Figueiredo et al., 2017; Hunsucker, 1982).

This interpretation is further supported by Han et al. (2025), who reported equatorward-propagating LSTIDs during the 10 May 2024 superstorm extending from mid-latitudes to low latitudes in both hemispheres. Although data gaps existed at higher latitudes, the authors identified multiple LSTIDs in the American sector, including disturbances exhibiting cross-polar propagation from the dayside to the nightside. Regarding structure 3, its negative ΔTEC signature is weaker in both sectors and appears more pronounced in the Peruvian sector than in the Brazilian sector, consistent with the oscillatory behavior of foF2 observed over JC between approximately 22:00 and 24:00 UT. In agreement with the keograms, this structure does not appear to propagate across the equator into the Southern Hemisphere as clearly as structure 2.

Additionally, LSTID modulation is evident in hmF2 over BL (Figure 2, between 18:30 and 22:30 UT), as indicated by its oscillatory behavior. However, over Jicamarca, the effects of LSTIDs on both foF2 and hmF2 appear less pronounced than at the Brazilian stations. This contrasting behavior suggests that, although the LSTID affected both regions, its impact on hmF2 was strongly modulated by local electrodynamic conditions. Over Brazil, the combined action of the LSTID, eastward PPEF, and favorable neutral winds likely promoted vertical plasma uplift, leading to variations in both hmF2 and foF2. In contrast, over the Peruvian sector, where the magnetic declination is close to zero, the F-layer height is primarily controlled by zonal electric fields through $E \times B$ drifts. It is possible that the large magnetic declination (~20°W) over Brazil introduces stronger coupling between zonal and meridional wind components, enhancing vertical plasma transport. As a result, the weaker or differently phased electrodynamic forcing over Jicamarca limited vertical displacement, leading predominantly to density variations (foF2) with minimal changes in hmF2 caused by the LSTIDs.

Based on these results, we suggest that the discrepancies between the SUPIM-INPE simulations in Figure 5 and the observations after 20:30 UT are likely associated with the effects of these large-scale propagating disturbances, which are not fully captured by the model.

This finding underscores the importance of considering the role of LSTIDs when interpreting ionospheric variability during disturbed periods, particularly in equatorial regions, where the coupling between electrodynamical processes and atmospheric disturbances is especially complex. Storm-time LSTIDs have been widely reported in the literature as important drivers of ionospheric uplift and variability during geomagnetic storms (e.g., Borries et al., 2009; Fesen et al., 1989; Prölss & Jung, 1978; Rishbeth, 1975). In addition, transequatorial TIDs have been observed during geomagnetic storms, producing longitudinal structuring in the ionosphere over the Brazilian sector (e.g., Batista et al., 2006; Bravo et al., 2019).

However, the presence of LSTIDs introduces an important limitation in the interpretation of the results shown in Figure 5 (Case 1). Since LSTIDs can significantly modulate neutral winds, the vertical plasma drift derived from Digisonde observations, which is used as a proxy for the zonal electric field, may include contributions from neutral wind effects. As a result, the inferred drift may not represent purely electrodynamic forcing, potentially masking the actual role of the background electric field.

To minimize this ambiguity and better isolate the electrodynamic contribution, an additional model run was performed considering only the zonal electric field, as described in the following paragraphs.

3.3.3. Vertical Plasma Drift Model for Brazil—Case 2

To avoid contamination by LSTIDs in the estimation of the vertical plasma drift and thereby isolate the electric-field effect, from 18:50 UT, the vertical drift over JC was adopted as the reference input from 18:50 UT onward, rather than the drift derived from ionosonde measurements. This choice was made because LSTID-related fluctuations are not detected by the Jicamarca radar measurements. As shown in Figure 8, the drift in this case was estimated using the F08 model between 00:00 and 16:50 UT, magnetometer-derived data between 16:50 UT and 18:50 UT, and the JC vertical drift variation from 18:50 UT onward ($F_{08}09May + \Delta Vz_{JC}$).

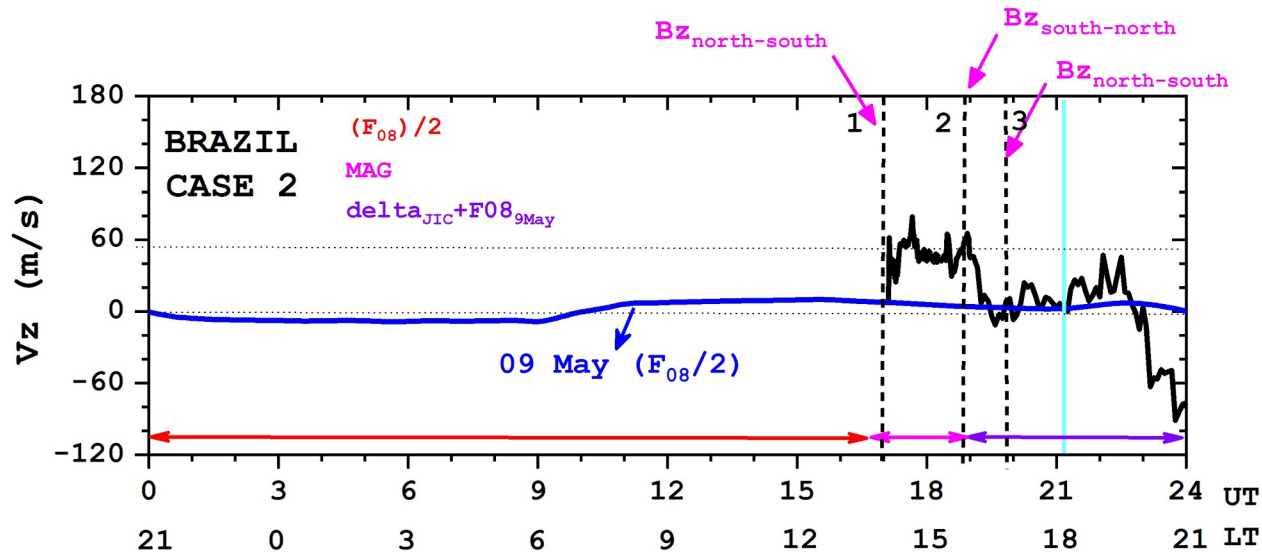


Figure 8. Second model of vertical plasma drifts on 10 May 2024 over Brazil used as input parameters in the SUPIM model. Blue curves indicate quiet time reference day (9 May). The colored arrows indicate the interval in which the drift values were obtained using measurements and the model. For further details, see the text.

3.4. SUPIM-INPE Results—Case 2

Figure 9 presents the SUPIM-INPE results considering only the effects of the electric field. In this case, the same methodology used to apply wind corrections in Figures 5 and 6 was adopted. Notable modifications in the hmF2 parameter appear after 19:00 UT; however, these results indicate that the electric field alone plays only a secondary role in the pronounced uplift of the F layer observed at this time.

Significant changes in foF2 are also evident after 22:30 UT. It is important to note that when the effects of LSTIDs on the vertical drift are removed, the discrepancies between the modeled and observed foF2 tend to become even larger. This degradation in model performance strongly suggests that electric-field-driven processes alone are insufficient to reproduce the observed ionospheric response.

Based on this analysis, together with the results discussed in Case 1, we conclude that LSTIDs were the dominant mechanism responsible for the strong uplift of the F layer at sunset and the associated modulation of foF2 over the Brazilian sector. This finding highlights the critical role of storm-time neutral atmosphere–ionosphere coupling and motivates further investigation using additional events and coupled modeling approaches.

Finally, Figure 9 also provides information on the F3 layer formation in the regions studied. Our results show that, for both BL and SL, the SUPIM-INPE model overestimates the maximum hmF2 values near 19:00 UT, even after the application of wind adjustments. This discrepancy can be related to the occurrence of the F3 layer, which was detected by the Digisonde at both stations, as illustrated by the ionogram sequence in Figure 10 and highlighted by the gray shaded area in Figure 9. Both BL (Figure 10a) and SL (Figure 10b), show signatures of two distinct layers from 17:50 UT, with the initial virtual height of the upper layer located around 600 km, which indicated the clear formation of the F3 layer. In agreement with Balan et al. (1998), three main factors are required for the formation of the F3 layer: (a) an upward plasma drift driven by an eastward electric field, (b) a meridional wind directed toward the equator, and (c) the usual photochemical and dynamical processes that generate the regular F layer. The F3 layer develops in regions where the combined effects of $E \times B$ drift, diffusion along magnetic field lines, and meridional winds result in a net upward motion.

Figure 10 also shows that over SL, the F3 layer persisted for 1 hr and 20 min (between 17:50 UT and 19:10 UT), while over BL, the F3 was registered between 17:50 UT and 18:40, therefore, with a shorter duration when compared to SL (note that, in Figure 10, only the ionograms displaying the initial and final F3 layer occurrence are shown). It can be noted that the F3 layer reached virtual heights of ~1,000 km over BL (18:40 UT) and 1100 km over SL (19:10 UT). Over JC, the ionogram sequence in Figure 3 also shows the presence of the F3 layer between 17:38 UT and 17:53 UT. However, for this station, it was not possible to determine the total duration of this layer due to the sounding limitation of the VIPIR. It is important to mention that in the presence of the F3 layer, the

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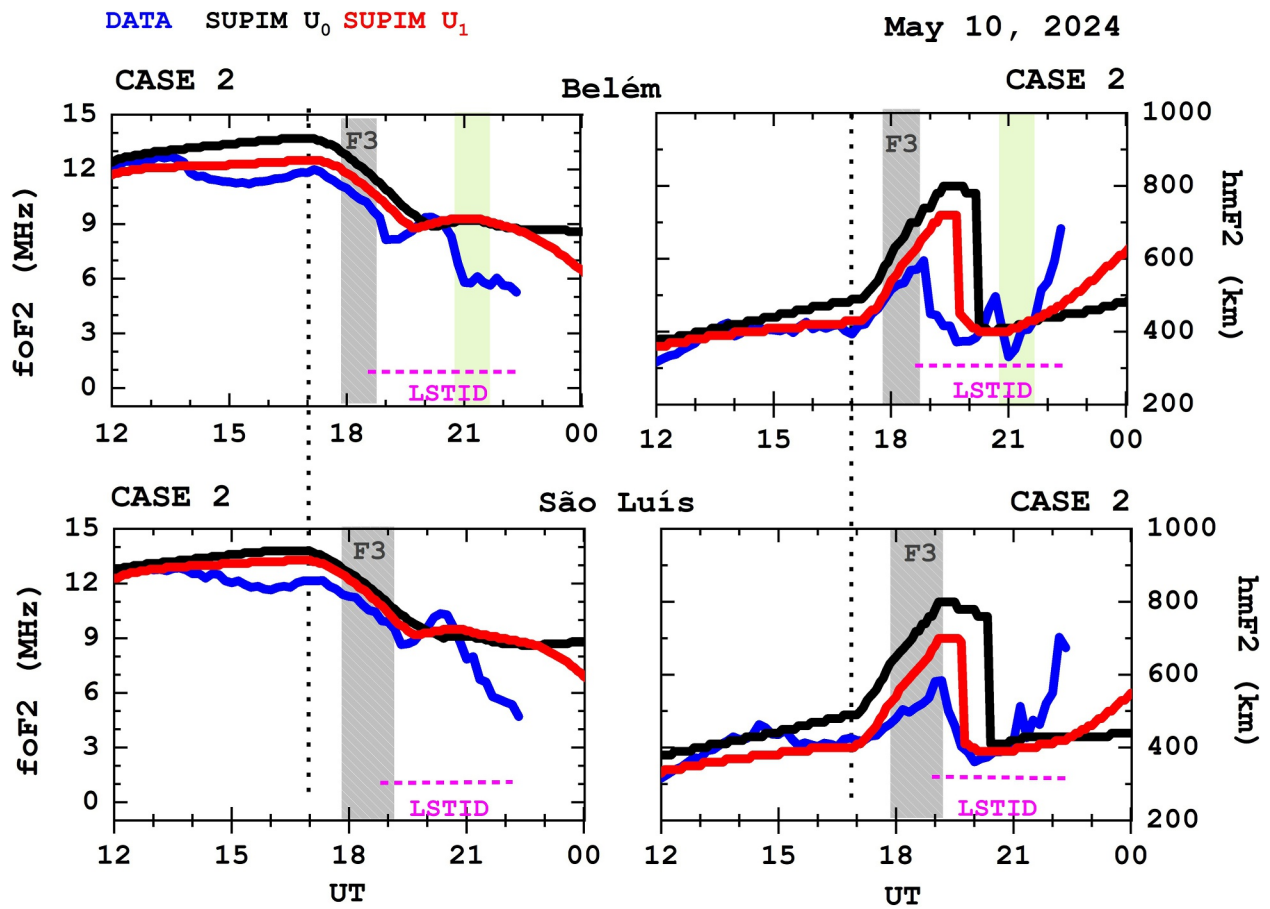


Figure 9. Model results for Belém (upper panels) and São Luís (lower panels) considering only the effects of electric fields (Case 2). The occurrence of the F3 layer and large-scale traveling ionospheric disturbances is also indicated in the panels. The pink dashed line indicates important fluctuation in foF2 and hmF2 between 18:30 UT and 22:30 UT, while the green shaded area over BL denotes intervals assigned with the URSI qualifying letter D (see text for details).

SAO-Explorer software cannot reliably determine the hmF2 parameter, which can likely explain the discrepancies between the observed and modeled data during this interval. This problem has already been reported previously (see e.g. Bravo et al., 2017).

Figure 11 shows the sounding frequency profiles as a function of height, comparing SAO-Explorer calculations with the results generated by SUPIM-INPE (Case 2) using the original (U_0) and modified (U_1) winds at 15:00 UT, 18:30 UT, 19:30 UT, and 20:30 UT, covering periods before, during, and after the peak in hmF2. Please note that the topside profile is represented by the dashed blue curve because it is calculated by a model in the SAO-Explorer software (Reinisch et al., 2009). At 15:00 UT, the wind correction primarily improved the foF2 results, with the modeled profiles closely matching the observations. This remarkable reproduction of the observed vertical profile by the model, using the modified wind U_1, gives us confidence that this approximation is appropriate for the event being simulated.

By 18:30 UT, the SUPIM profiles for both U_0 and U_1 show a deformation relative to the previous state, exhibiting a tendency to form two distinct peaks, one at lower altitudes and another at higher altitudes, indicating the incipient formation of the F3 layer. This double-peak structure becomes more pronounced at 19:30 UT, particularly when U_0 is used as input, which shows a broader profile with a higher peak around 800 km. When U_1 is used, the profile also broadens, but the more intense peak is located near 400 km. By 20:30 UT, the double-peak structure in the SUPIM profile using U_1 has smoothed, whereas it persists in the simulation using U_0, suggesting that the widest hmF2 peak observed in this case (between 17:00 UT and 21:00 UT, Figure 5) is due to the model continuing to reproduce the F3 layer.

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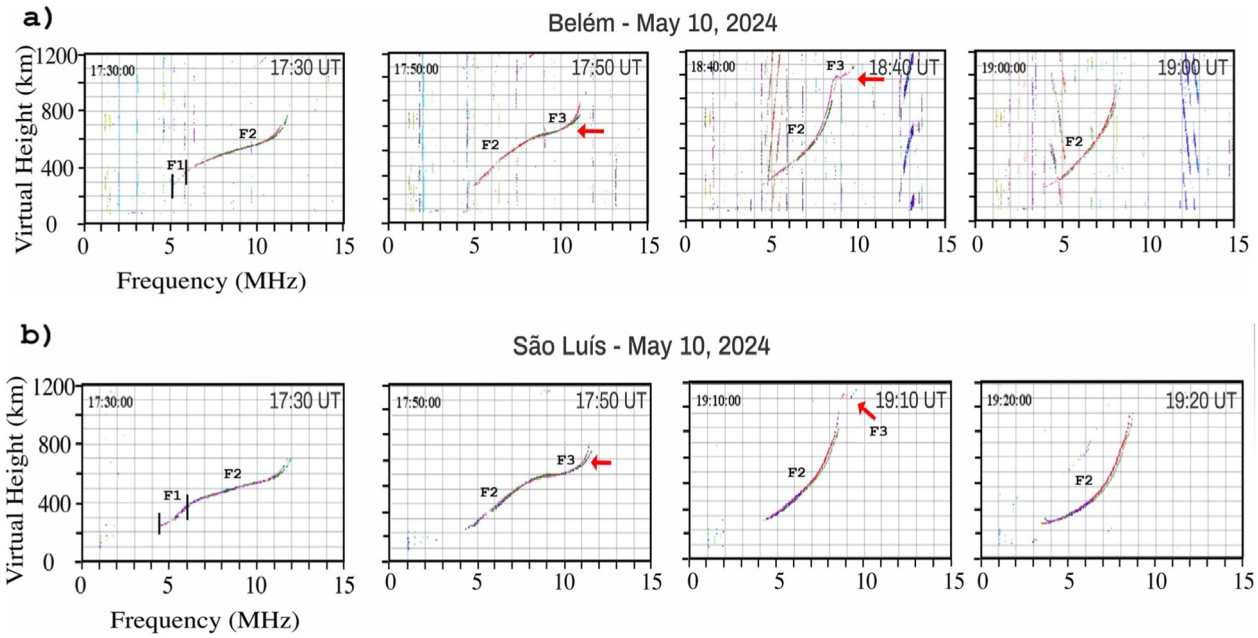


Figure 10. Ionograms' sequences over (a) Belém and (b) São Luis showing the F3 layer formation on 10 May 2024.

A noteworthy aspect is that, as mentioned previously, the wind favoring F3 layer formation is directed toward the equator, as it generates an upward plasma drift. In the case of U_0, this condition is satisfied because the wind over the Brazilian sector is northward during the period under discussion. However, the correction applied to U_0 produced a strong southward wind (U_1), which is opposite to the direction required for F3 layer formation according to Balan et al. (1998). Nevertheless, it is important to notice that the results from SUPIM-INPE are affected by the wind integrated along the magnetic field lines due to nonlinear effects, while the wind shown in Figure 6 is for a specific region. The resultant vertical plasma motion depends on the combined effect of the electric field and the wind. This upward component is also influenced by the geometry of the magnetic field and may therefore occur either directly above the station or in nearby regions. During the studied event, characterized by strong storm-time electric fields, this coupling between electric field–driven E × B drifts and wind-driven plasma transport makes the formation of an F3 layer with electron density exceeding that of the F2 layer particularly complex.

4. Conclusions

This paper discusses the equatorial ionospheric response over the American sector to the extreme geomagnetic storm of 10 May 2024. To this end, observational data from the Brazilian and Peruvian sectors were analyzed in conjunction with the SUPIM-INPE model. The main findings of this study are summarized below:

- A simultaneous and abrupt rise of the F2 layer was observed over SL, Belém, and Jicamarca during the onset of the storm's main phase. This enhancement was well reproduced by the SUPIM-INPE model, indicating that although the primary contribution originated from the eastward disturbed electric field, the modification in hmF2 over Brazil was also influenced by meridional winds.
- The complex descend phase of hmF2 near 19:00 UT seems to be controlled by a combination of factors, such as the overshielding electric field, meridional winds, F3 layer and LSTIDs. Over SL, the effect of the wind, for example, was more evident when compared to BL.
- The SUPIM-INPE model can reproduce the occurrence of the F3 layer over Brazil under both configurations considering and not considering wind corrections. However, the model overestimates the duration of this layer when compared with the Digisonde observations. Since the analyzed event took place under strong storm-time electric field disturbances, the contribution of the wind to F3 layer formation becomes more complex and difficult to isolate. Over JC, the F3 layer was identified earlier than over BL and SL.

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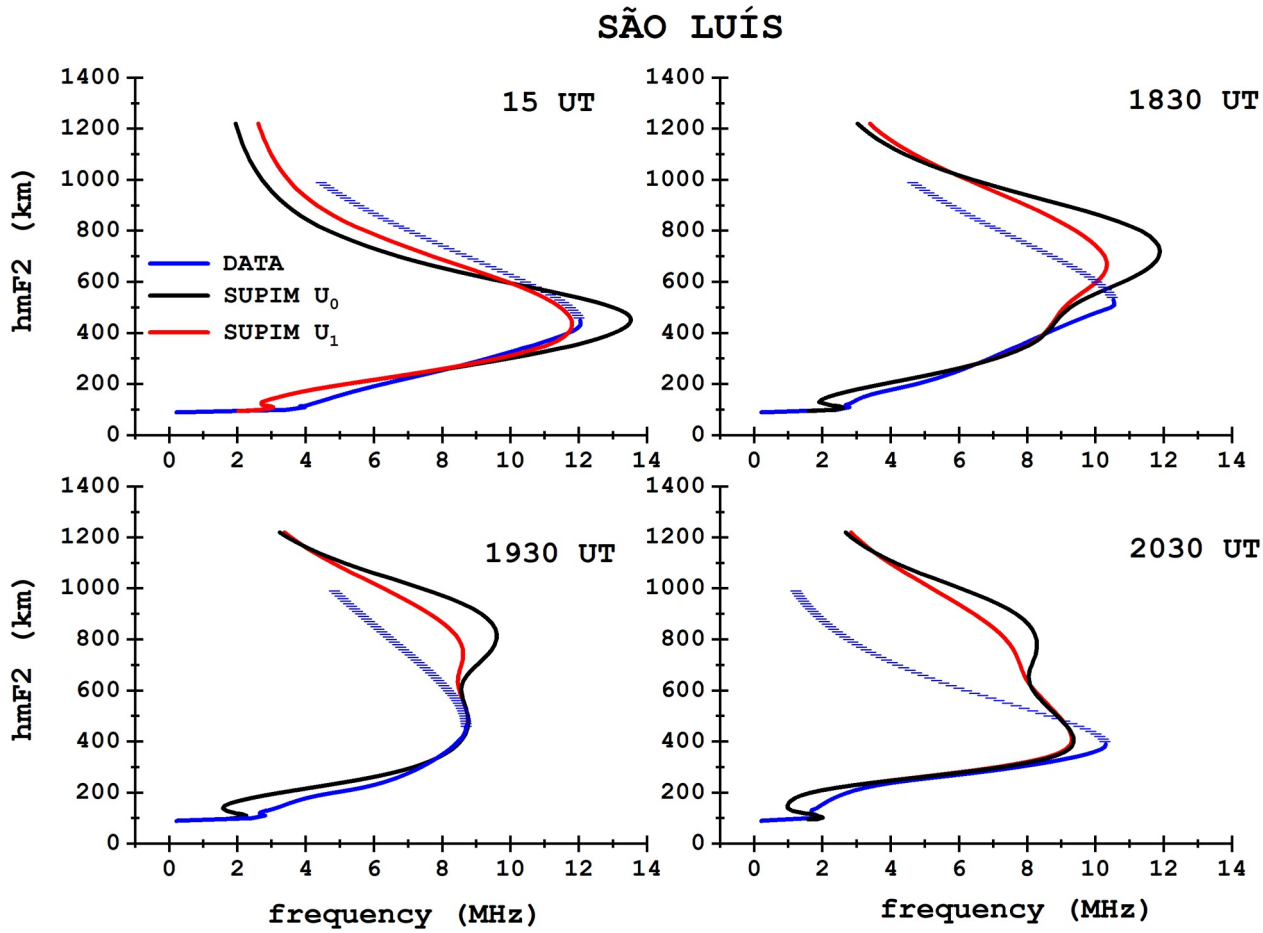


Figure 11. Variation of frequency profile based on data (blue) and calculated using the SUPIM-INPE model (black and red) for different time. The model calculations were performed using the drift described in Case 2.

- Around and after 20:00 UT, following the southward turning of Bz, a strong rise in the F layer was observed over Brazil, which was followed by a strong decrease in foF2. However, over JC, such features were not evident as in Brazilian regions. As indicated by the SUPIM-INPE model (Case 2), the more pronounced response over Brazil may be a result of the more effective contribution of the LSTID, which, coincidently combined with the typical enhancement of the zonal electric field at sunset, caused the F layer to reach unusually high altitudes.
- Besides the fact that JC is located closer to the magnetic equator than the Brazilian stations, the more pronounced influence of LSTIDs over Brazil compared to Peru may also be related to the large magnetic declination angles at the Brazilian stations. Under these conditions, meridional neutral winds associated with LSTIDs can develop a larger effective component along the geomagnetic field lines, thereby enhancing the wind-driven dynamo electric fields and modifying the local $E \times B$ plasma drifts. This mechanism may increase the efficiency with which LSTID-induced neutral wind perturbations modulate the ionospheric electrodynamics over Brazil. However, this hypothesis requires further investigation using coordinated observations and numerical modeling.
- In future work, the incorporation of observational wind data into the model is expected to help better constrain the parameters driving the ionospheric variability over the Brazilian sector. Although this approach may not fully resolve all the discrepancies between model and data, given that the model operates along magnetic field lines rather than at a single location, it should provide a more realistic representation of storm-time dynamics.
- Finally, the study of the daytime equatorial ionosphere using Digisonde data is particularly challenging because during the daytime the ionosphere becomes highly conductive due to the presence of the E region. This enhanced conductivity tends to short-circuit penetrating electric fields, reducing their effectiveness in

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producing vertical plasma drifts. Consequently, identifying the F-layer uplift associated with these disturbed electric fields becomes more difficult. Despite these limitations, the combined use of multi-instrument observations and modeling in this study allowed us to clearly identify and interpret the ionospheric response during this complex event.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

Digisonde and magnetometer data over Brazilian stations are available for download upon registration through the Embrace webpage of INPE's Space Weather program at http://www2.inpe.br/climaespacial/portal/en. In the case of Jicamarca, the JULIA radar and VIPIR ionosonde data can be obtained from the LISN Database (http://lisn.igp.gob.pe/jdata). The interplanetary medium conditions and geomagnetic indices were obtained from the OMNIWeb database (https://omniweb.gsfc.nasa.gov/form/omni_min.html).

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